Waseem Raza, PhD

Machine Learning Engineer | Applied AI & Data Science

Applied Machine Learning | Data Science | Model Evaluation | Intelligent Automation

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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Profile

Machine Learning Engineer and Data Scientist with a PhD in Electrical and Computer Engineering and 7+ years of combined industry and research experience developing predictive models, reliable ML workflows, and scalable analytics systems. Skilled in Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost, classification, regression, anomaly detection, time-series forecasting, domain adaptation, feature engineering, cross-validation, data quality, Azure Databricks, Apache Spark, MLOps, and model evaluation. Experienced in converting complex operational data into validated decision-support systems.

Technical Competencies

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, transfer learning, domain adaptation, adversarial learning, anomaly detection, classification, regression, time-series forecasting, cross-validation, and model evaluation

Data Engineering: ETL/ELT pipelines, AI-ready datasets, batch processing, ingestion, cleansing, schema transformation, validation, feature engineering, data-quality assessment, and structured and unstructured data workflows

Data & AI Platforms: Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Azure Synapse Analytics, Azure Functions, Azure Blob Storage, Azure Data Lake, Google BigQuery, and cloud analytics workflows

Programming & Analytics: Python, SQL, C/C++, and R; Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Plotly, Streamlit, Jupyter, and exploratory data analysis

Generative & Agentic AI: LLM applications, retrieval-augmented generation (RAG), prompt engineering, document ingestion, chunking, embeddings, semantic retrieval, structured information extraction, tool-using agents, agent orchestration, and grounded response generation

LLM Frameworks & Models: Hugging Face Transformers, LangChain, LangGraph, OpenAI APIs, transformer architectures, fine-tuning concepts, NLP foundation models, prompt templates, structured outputs, and LLM evaluation

MLOps & Software Engineering: Docker, Kubernetes, OpenShift, Git, GitHub Actions, CI/CD, REST APIs, Flask, testing, Linux, shell scripting, model deployment, reproducible workflows, and cloud-native application development

Microsoft AI Ecosystem: Microsoft Foundry, Azure OpenAI, Azure AI Services, Azure AI Search, Azure Machine Learning, Foundry models and agents, Azure Content Understanding, Semantic Kernel, and responsible AI concepts

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer **02/2025 - Present**

AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Own end-to-end development, testing, and operational integration of machine-learning diagnostics for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Design domain-adaptive neural models and reusable preprocessing pipelines that improve generalization across heterogeneous operational domains and distribution shifts.
- Build model-validation, cross-domain evaluation, error-analysis, and failure-mode assessment procedures to support reliable production deployment.
- Translate network and public-safety requirements into measurable ML objectives, validated data assets, and production-oriented decision-support workflows.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**

Impac Exploration Services *Weatherford, OK*

- Built machine-learning and LLM-assisted analytical workflows for large geochemical datasets using Python and Azure cloud services.
- Developed scalable ingestion, cleansing, feature-preparation, transformation, and validation workflows using Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Graduate Research Assistant **09/2019 - 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Projects & Research

Projects & Applied Research

Financial ML and Forecasting *Portfolio Project | 2024-2025*
Developed finance-oriented machine-learning workflows for credit-risk and loan-default modeling together with time-series forecasting experiments. Applied data preparation, feature engineering, model comparison, cross-validation, and predictive-performance analysis to structured financial datasets.
Tools & Methods: Python, scikit-learn, Classification, Credit Risk, Loan Default Modeling, Time-Series Forecasting, Feature Engineering, Cross-Validation, Model Evaluation

Wearable Intelligence for Health Monitoring *Research Project | 2024-2026*
Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.
Tools & Methods: Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

Cloud Computer Vision (GCP AutoML / Object Detection) *Guided Project | 2025*
Implemented cloud-based computer-vision workflows for image classification, object detection, and vehicle counting using managed model services and Python inference pipelines. Evaluated preprocessing, inference outputs, class/count aggregation, and model performance across visual-data tasks.
Tools & Methods: Python, Computer Vision, Object Detection, Image Classification, Google Cloud AutoML, Streaming, Preprocessing, Inference, Model Evaluation

AI-Ready Lakehouse Pipeline *Portfolio Project | 2024*
Designed an AI-ready lakehouse pipeline that prepares structured and semi-structured data for analytics and downstream machine-learning workloads. Implemented repeatable ingestion, cleansing, transformation, validation, and scalable processing patterns suitable for reproducible cloud data workflows.
Tools & Methods: Azure Databricks, Apache Spark, PySpark, ETL/ELT, Data Ingestion, Schema Transformation, Data Validation, Feature

Preparation, Cloud Data Engineering

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

OCR & Document Intelligence Extraction Pipeline *Guided Project | 2025*

Implemented a document-intelligence pipeline that converts reports, invoices, and contract-style documents into structured outputs through OCR, layout/table parsing, entity and field extraction, normalization, and validation. Designed the workflow for downstream analytics and data integration rather than isolated text extraction.

Tools & Methods: OCR, Document Intelligence, LLMs, Named Entity Recognition, Table Extraction, Structured JSON, Normalization, Validation, Data Integration

Agentic AI and RAG Applications *Portfolio Project | 2026*

Built grounded GenAI workflows that combine document retrieval, vector search, stateful orchestration, tool-using agents, memory, evaluation, and multi-agent handoffs. Emphasized controllable execution, traceable context flow, and reliable responses for multi-step tasks rather than single unconstrained model calls.

Tools & Methods: Python, LangGraph, LangChain, CrewAI, AutoGen, RAG, Vector Databases, Embeddings, Tool Calling, Agent Memory, Evaluation

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *2026*
HealthINF | Accepted

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *2025*
Sensors 2025

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *2024*
IEEE PIMRC

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *2024*
IEEE SmartNets

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *2022*
IEEE WCNC

Certifications & Professional Development

1. Machine Learning — Stanford Online / Coursera | Completed | March 29, 2020

Focus: Covers foundational machine-learning methods for supervised and unsupervised problems, including regression, classification, model evaluation, regularization, clustering, and practical model-selection considerations. Emphasizes translating data into predictive models while controlling generalization error.

Tools & Skills: supervised learning, regression, classification, regularization, clustering, model evaluation, bias-variance tradeoffs, machine-learning workflows

2. Introduction to Data Analytics in Google Cloud — Coursera | Completed | July 24, 2024

Focus: Introduces cloud-native data analytics in Google Cloud, covering the analytics lifecycle, data sources, storage and processing concepts, and how cloud services support scalable analysis. Emphasizes organizing data work to produce reliable analytical insight.

Tools & Skills: Google Cloud, cloud analytics, data lifecycle, data-ingestion concepts, storage and processing, analytical workflows, data-driven decision support

3. Generative AI Engineering and Fine-Tuning Transformers — IBM / Coursera | Completed | August 6, 2026

Focus: Covers transformer-based generative-AI engineering, including pretrained language models, model adaptation, fine-tuning strategies, and evaluation. Emphasizes converting reusable foundation models into task-focused systems while understanding training and inference tradeoffs.

Tools & Skills: transformers, foundation models, fine-tuning, transfer learning, tokenization concepts, model adaptation, LLM evaluation

4. Introduction to Deep Learning & Neural Networks with Keras — IBM / Coursera | Completed | August 12, 2026

Focus: Introduces deep-learning and neural-network fundamentals using Keras, covering network architecture, training, validation, and model improvement. Connects concepts such as activation functions and optimization with practical neural-network implementation and evaluation.

Tools & Skills: Keras, deep learning, neural networks, model training, validation, activation functions, optimization, model evaluation

5. Python for Data Science, AI & Development — IBM / Coursera | Completed | July 12, 2026 | Certificate

Focus: Builds practical Python skills for data science and AI development, including core programming, data structures, data manipulation, and application-oriented analytical workflows. Emphasizes using Python as a reusable foundation for data processing, modeling, and AI applications.

Tools & Skills: Python, data structures, data manipulation, Pandas concepts, NumPy concepts, data-analysis workflows, AI application development